INFOTACT SOLUTIONS — FORMAL INTERNSHIP CAPSTONE

EconoCausal: Dynamic Pricing via Double Machine Learning

Heterogeneous Treatment Effect Estimation & Prescriptive Marketing Budget Allocation Under Strict Capital Constraints

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2. Abstract

Promotional discounting represents one of the largest capital outlays in modern retail and e-commerce, yet traditional predictive machine learning models systematically misallocate promotional spend. Supervised learning algorithms predict customer conversion probabilities conditional on treatment, mistaking correlation for causation: loyal, high-intent shoppers receive the steepest discounts simply because they purchase frequently, creating massive deadweight loss (subsidizing purchases that would have occurred at full price).

EconoCausal solves this fundamental econometrics challenge by coupling Double Machine Learning (DML) with an exact Mixed-Integer Linear Programming (MILP) budget solver. Using EconML LinearDML with 5-fold cross-fitting and LightGBM nuisance models, the system isolates true Individual Treatment Effects (CATE / τ■■) across multiple promotional discount tiers (■0, ■10, ■20). These price elasticities feed into a SciPy HiGHS optimization engine that prescriptively allocates discount offers to maximize total incremental profit under a strict ■50,000 marketing budget cap.

Across a 10,000-customer benchmark cohort, EconoCausal generates **■3,76,115.07 in net incremental profit (7.52x ROAS)**, outperforming random promotional rollouts by **+75.5% (+■1,61,765.07)** while leaving 5,432 organic buyers at full price. The causal model satisfies all five reported causal falsification diagnostics (including Placebo p=0.485, Confounder drift 0.16%, Subset drift 0.03%, Positivity drift 0.17%, and the Model-Based Residual Permutation Diagnostic p=0.0020) and achieves post-matching covariate balance (|SMD| < 0.022 across all 9 confounders). The complete solution is delivered via an asynchronous FastAPI microservice and an interactive React 19 executive dashboard.

3. Problem Statement & Business Motivation

In enterprise retail transactional systems, historical discount assignments are rarely randomized. Marketers preferentially offer coupons to cart abandoners, VIP loyalty members, and high-frequency shoppers. Consequently, customer covariates X simultaneously influence both treatment assignment T and purchase outcome Y, inducing severe confounding selection bias:

Causal Identification Gap: $E[Y \mid T=1] - E[Y \mid T=0] \neq E[Y(1) - Y(0)]$

Standard supervised models optimize associative loss functions $\text{argmin}_\theta L(Y, f_\theta(X, T))$. When deployed operationally, these models recommend discounting customers who already exhibit high baseline conversion propensities, wasting marketing capital and eroding operating margins.

4. Objectives & Four Modular Pillars (Infotact Syllabus)

- **Module 1 (Causal Graphing):** Microsoft DoWhy DAG formulation identifying 9 observable confounders, treatment tiers, and purchase outcomes.
- **Module 2 (Double ML Engine):** EconML LinearDML with LightGBM base estimators to residualize confounding and predict customer CATE elasticities.
- **Module 3 (Prescriptive Optimization):** Multiple-Choice Knapsack problem in SciPy HiGHS MILP maximizing net profit under nominal and redeemed budget caps.
- **Module 4 (Full-Stack Dashboard):** High-performance React 19 + Plotly interface backed by a FastAPI REST API with automated customer behavior drift detection (PSI).

5. Dataset Schema & Anti-Leakage Safeguards

The cohort comprises N = 10,000 customer transaction records. Features include 9 verified pre-treatment confounders: age, monthly_income_inr, web_visits_last_30d, past_purchases_count, days_since_last_purchase, cart_baseline_value_inr, loyalty_member, device_Mobile, device_Tablet.

Strict Anti-Leakage Guarantees: Post-treatment variable 'discount_depth_ratio' is strictly excluded from pre-treatment covariate matrices. True counterfactual outcomes Y(0), Y(10), Y(20) are isolated in ground_truth_eval.csv and never exposed to the optimizer or operational API endpoints.

6. Quantitative Benchmark Results (■50,000 Budget Cap)

Policy Architecture	Budget Spent	Targeted Shoppers	Inc. Orders	Net Incremental Profit	RODS
DML MILP Optimal Policy	■50,000.00	4,568 (45.7%)	+829.88	■3,76,115.07	7.52x
Random Allocation Heuristic	■50,000.00	5,000 (50.0%)	+414.94	■2,14,350.00	4.29x
High Cart-Value Heuristic	■50,000.00	5,000 (50.0%)	+584.20	■2,92,800.00	5.86x
Flat Blanket Heuristic	■50,000.00	5,000 (50.0%)	+414.94	■2,14,350.00	4.29x
Zero Discount Control	■0.00	0 (0.0%)	0.00	■0.00	0.00x

7. Econometric Falsification & Covariate Balance

- **Placebo Treatment Permutation:** p = 0.485 (Effect collapsed from +14.22% to +0.04% under random shuffling; PASS).
- **Random Confounder Injection:** 0.16% relative effect drift under standard normal noise (< 5.0% tolerance; PASS).
- **Data Subset Stability:** 0.03% relative drift across 10 bootstrap 80% sub-samples (< 6.0% tolerance; PASS).
- **Positivity Trimming:** 0.17% relative drift after trimming extreme tails (< 8.0% tolerance; PASS).
- **Residual Permutation Diagnostic:** Permutation uplift +14.22% vs null range [-2.12%, +2.02%] (p = 0.0020; PASS).
- **Covariate Balance (PSM):** 1:1 Nearest-Neighbor Caliper Matching achieved |SMD| < 0.022 across all 9 confounders (94%+ bias reduction).
- **Quantitative Sensitivity:** Rosenbaum bounds Γ* = 1.7 (■10 arm), 2.1 (■20 arm); Cinelli-Hazlett RV = 12.4%, 17.8%.

8. Automated Test Execution & Quality Gate

The codebase defines 66 distinct automated unit tests (64 safe runnable, with 2 permanently quarantined against benchmark data leakage):

- 02_Causal_ML/test_dml.py: 5 tests (test_05 passed live; test_01, test_02 WDAC-constrained; test_03, test_04 quarantined)

- 03_Backend_Optimization/test_optimizer.py: 13/13 LIVE PASS (23.8s, 100%)
- 03_Backend_Optimization/test_validation.py: 14 Tests (Historical 100% Pass; Host WDAC Constrained in L8)
- 04_Frontend_Dashboard/backend/test_api.py: 34 Tests (33 LIVE PASS; test_19 WDAC-constrained)
- Technical Checks: Dataset CLI (Pass), Vite Build (Pass 2.22s), Linter (WDAC Blocked), Native Print (Not Verified)

9. Conclusion & Deployment Recommendations

EconoCausal proves that combining Double Machine Learning with Mixed-Integer Linear Programming transforms promotional discounting from an uncontrolled cost center into a prescriptively optimized profit driver. For enterprise live deployment, we recommend initiating a 10% randomized A/B validation test before fully scaling automated discount allocations.

Report Certified by: Vishal Yadav (Data Associate L1. Intern) | Infotact Solutions | October 2026